

Week 4, Day 1 — presenter notes

Slides: open [week-04-day-01.html](#) → **F** fullscreen → arrows. **N** notes. **P** print/PDF.

This session is **modules, packages, layout, SMS kickoff**. No HTTP. No Gin. No database. The product clock starts.

Timing (3–3.5 hours)

Clock	Block	Slides
0:00–0:15	Week 3 homework check	1–2
0:15–0:30	Outcomes + product clock	3–4
0:30–1:00	Packages, imports, export, <code>internal/</code>	5–8
1:00–1:25	Modules, <code>go.mod</code> , tidy (light)	9–11
1:25–2:10	Layout, <code>cmd/server</code> , package jobs (live scaffold)	12–15
2:10–2:35	Clean code, README, PR, <code>.env</code>	16–19
2:35–end	Lab: skeleton + PR URL	20–end

Prefer a finished skeleton over a perfect lecture on `go get`. Never skip `.gitignore` for `.env`.

Live demo script

```
mkdir -p ~/epic-go/sms/backend
cd ~/epic-go/sms/backend
go mod init student-management-system
mkdir -p cmd/server
internal/{config,handlers,repository,models,routes,middleware,database,utils}
```

`cmd/server/main.go`:

```
package main

import "fmt"

func main() {
```

```
fmt.Println("sms starting")
}
```

```
gofmt -w cmd/server/main.go
go run ./cmd/server
```

Track empty dirs with `.gitkeep` files so Git keeps them.

Lab gate

- [] `go.mod` module `student-management-system`
- [] Folder tree matches the course layout
- [] `go run ./cmd/server` → `sms starting`
- [] README: three product sentences + how to run
- [] `.env` gitignored; no binaries
- [] PR URL collected

Homework

Same gate, as a GitHub PR. No Gin yet.

Typical failures

Symptom	Fix
<code>package ... is not in std</code>	missing / wrong <code>go.mod</code>
<code>go run main.go</code> from repo root	<code>go run ./cmd/server</code> from <code>backend/</code>
empty folders missing on GitHub	add <code>.gitkeep</code>
<code>.env</code> in the PR	remove; fix <code>.gitignore</code>
module path typos in future imports	keep <code>student-management-system</code> consistent

What not to teach today

`net/http`, Gin, JSON handlers, PostgreSQL, JWT, structs beyond naming folders, interfaces. Tease week 5 only.